

Practice exercise, Lecture 3

ØKA2015 Applied Data Science. Not graded. Do it before Friday; it is the warm-up for Assignment 1.

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Reference: the Lecture 3 slides and `lecture03_tidyverse.R`. Use the `students.csv` file, either from the web, `read_csv("https://pos.it/r4ds-students-csv")`, or from `data/students.csv` if you opened the lecture project.

Work in a script. For each task, write the command, run it with **Ctrl + Enter**, and check the Console. Six tasks, about 45 minutes.

Task 1: objects and vectors

The coffee bar sold 18, 22 and 27 lattes on Monday, Tuesday and Wednesday, at 52 NOK each.

- Store the three counts in a vector called `cups`, and the price in `price`.
- Compute the revenue for each day in one line, without a loop.
- Compute the total revenue, and the average number of cups per day.
- What does `cups > 20` return? Explain each element of the answer.

Task 2: packages

- Of these two lines, which do you run once per computer, and which every session?

```
install.packages("tidyverse")  
library(tidyverse)
```

- You run `library(janitor)` and get `there is no package called 'janitor'`. Write the single line that fixes it.

Task 3: reading data

- Read the students file into an object called `students`.
- Show the first six rows.
- How many observations and how many variables does it have? Use one function for each.
- What does `glimpse(students)` show you that `head(students)` does not?

Task 4: types

- Look at the type printed under the `AGE` column. Why is it `<chr>` and not `<dbl>`?
- Look at `favourite.food` for student 3. Is that a missing value to R? How do you know?
- Read the file again so that "N/A" becomes a real `NA`. Then count the missing cells.

Task 5: a sentence with numbers in it

Using `nrow()`, `ncol()` and `cat()`, print exactly this line to the Console:

The students table has 6 rows and 5 columns.

Task 6: save your work

Put every command you used into a script, with a comment above each one saying what it does, and save it as `YourName_lecture03.R`. Close RStudio, open it again, and run the whole script from the top. It should work without you typing anything else. If it does not, the missing line is almost always `library(tidyverse)`.

Answers are not distributed. Compare with a classmate, or bring your script to Friday's lecture.